

Wireless GATE ALARM

PRADEEP G.

This wireless gate alarm uses infrared (IR) and 433MHz RF modules. IR transmitter unit emits 36kHz signal and is fitted on one side of the gate's frame. IR receiver unit and RF transmitter module, along with the encoder, are fitted on the opposite side of the gate's frame. The alarm unit is kept inside the house, and it uses 433MHz RF receiver module and musical alarm circuit.

Circuit and working

Circuit of the wireless gate alarm has three units, namely, IR transmitter unit, IR receiver unit and alarm unit. Modulated invisible IR beam is present between transmitter and receiver units installed at the gate.

Circuit diagram of the IR transmitter unit is shown in Fig. 1. By adjusting potmeter VR1, frequency can be set at 36kHz. At this frequency, maximum range is obtained (more than five metres).

IC NE555 (IC1) is wired as an astable multivibrator. Output signal from NE555 is amplified by npn transistor SL100 (T1) to drive two IR-transmitting LEDs (LED1 and LED2) in series.

Circuit diagram of the IR receiver unit is shown in Fig. 2. It is built around IR receiver module TSOP1736 (IR RX1), timer NE555 (IC2),

encoder HT12E (IC3), 433MHz RF transmitter module (TX1) and some other components.

Circuit diagram of the alarm unit is shown in Fig. 3. It is built around 433MHz RF receiver module (RX1), decoder HT12D (IC4), melody generator UM66 (IC5) and some other components.

Encoder IC3 encodes four-channel transmitting signals. The address pins A0 through A7 of IC3 are used for security codes. Here, all the address pins are grounded. Decoder IC4 decodes four-channel signals. IC3 has the same address pins as IC4.

When a person crosses the gate, IR beam is interrupted for a moment.

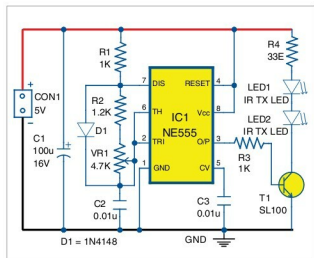


Fig. 1: Circuit diagram of the IR transmitter unit

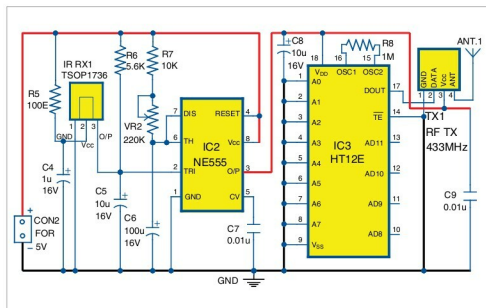


Fig. 2: Circuit diagram of the IR receiver unit

PARTS LIST

Semiconductors:

- IC1, IC2 - NE555 timer
- IC3 - HT12E encoder
- IC4 - HT12D decoder
- IC5 - UM66 musical melody
- T1 - SL100 npn transistor
- T2, T3 - BC547 npn transistor
- ZD1 - 3.3V Zener diode
- D1 - 1N4148 signal diode
- LED1, LED2 - IR transmitting LED

Resistors (all 1/4-watt, 5% carbon):

- R1, R3 - 1-kilo-ohm
- R2 - 1.2-kilo-ohm
- R4 - 33-ohm
- R5 - 100-ohm
- R6 - 5.6-kilo-ohm
- R7 - 10-kilo-ohm
- R8 - 1-mega-ohm
- R9 - 33-kilo-ohm
- R10 - 2.2-kilo-ohm
- R11, R12 - 680-ohm
- VR1 - 4.7-kilo-ohm potmeter
- VR2 - 220-kilo-ohm potmeter

Capacitors:

- C1, C6, C11 - 100uF, 16V electrolytic
- C2, C3, C7, C9 - 0.01uF ceramic disk
- C4, C10 - 1uF, 16V electrolytic
- C5, C8 - 10uF, 16V electrolytic

Miscellaneous:

- CON1-CON3 - 2-pin terminal connector
- IR RX1 - TSOP1736 IR receiver module or its equivalent
- TX1 - 433MHz RF transmitter
- RX1 - 433MHz RF receiver
- LS1 - 8-ohm, 0.5W loud speaker
- ANT.1, ANT.2 - 17cm single-strand hook-up wire antenna
- 5V DC power supply